Technical Task Brief

Intern Work Plan · 6 Weeks

Indian Institute of Technology Kharagpur

Duration 6-week engagement starting [date to be confirmed]

Reporting One-page written update submitted every Friday by 6 PM

Contact PI (contact details shared separately)

Repository Access granted on Day 1

Team at a Glance

#	Intern	Primary Role	Core Deliverables (6 Weeks)
1	**Sahil Kumar Gupta**	**Backend + Doctor/Admin Portal + RAG**	DB migration · Alert engine · Gemini RAG fallback · Doctor dashboard APIs · Admin portal · Cloud Run deployment
2	**Saugata Malakar**	**Healthcare Privacy + LLM + Multimodal AI**	Data anonymisation (DPDP Act 2023) · Federated learning PoC · Multimodal AI integration · Clinical NLP
3	**Adreesh Mitra**	**Wound Image Processing & Segmentation**	OpenCV preprocessing · SAM2 wound segmentation · Coin-based area measurement · Wound boundary AI
4	**Sharif Hossain Sarkar**	**Wound AI Model Training & Deployment**	Wagner Grade EfficientNet-B0 · Wound colour CNN · TFLite export · Flask inference endpoints
5	**Kousttav Paul**	**Skin AI + Mobile App Features**	EfficientNet-B3 skin model · Leprosy recall ≥95% · React Native screen fixes · ASHA training modules
6	**Shivraj Gulve**	**Eye AI + RAG + Risk Scoring + Deployment**	MobileNetV3 eye triage · Diabetic risk scoring · LlamaIndex RAG · pgvector · Streamlit analytics

Saugata Malakar | Healthcare Privacy · LLM · Multimodal AI

Your stack	Python, spaCy, LangChain, FAISS, Flower (FL), Gemini API, python-docx
You are responsible for	DPDP Act 2023 compliance, data anonymisation, federated learning PoC, multimodal AI integration, clinical NLP, RAG assistant for field workers, consent versioning, Privacy Impact Assessment
First-day priority	Get the PII field map done before any other work starts. Every other task depends on knowing which fields are sensitive.
Semester	Part of this, FL development and PIA updates, could continue part-time. Discuss with the PI.

What you are building

Your work is about trust. The platform handles sensitive medical data, and every piece of your work either makes it safer (anonymisation, encryption audit, consent versioning) or more useful (multimodal AI, clinical NLP, field worker assistant). None of the data pipeline can go live without your anonymisation module in place. The Privacy Impact Assessment is what the ethics committee reviews, if it is weak, the research track stalls.

Week by week

Week	Theme	What to do	What to hand in
Week 1	**PII audit & architecture**	• Go through the 26-table database schema and classify every column: direct identifier (name, phone, Aadhaar number), quasi-identifier (fields that in combination can re-identify someone), or non-sensitive. Document this in a PII map, it is the foundation for everything else you build • Write the DPDP Act 2023 compliance gap analysis. Three things are non-negotiable: all patient data stays in Mumbai (data localisation), patients can request deletion within a 72-hour window, and no raw patient images leave the platform unprocessed for external use • Design the anonymisation architecture on paper before writing code. Get it reviewed by the PI before Week 2	◦ PII field map (all 26 tables, three-tier classification) ◦ DPDP Act 2023 gap analysis ◦ Anonymisation architecture design (reviewed before Week 2 coding starts)
Week 2	**Anonymisation pipeline**	• Build the anonymisation module: pseudonymise patient IDs with HMAC-SHA256 and rotating salt, generalise age to 5-year bands, strip village names from exports. Target k-anonymity of at least 5, no record in an exported dataset should be uniquely re-identifiable • This module sits in the path of every data export. Nothing leaves the database for external analysis without passing through it • Work with the backend engineer on the patient erasure pipeline. He builds the API endpoint; you write the deletion logic that clears all 26 tables in correct dependency order. Test it, run the deletion, then verify nothing remains	◦ Anonymisation module (Python, unit tested, k≥5 verified) ◦ Erasure pipeline covering all 26 tables (run and verified) ◦ Data export endpoint integrated with the anonymisation layer
Week 3	**Federated learning PoC**	• Set up Flower (FL framework) with three simulated client nodes on localhost. Wrap the wound classification model as a Flower client. Run a federated round: each client trains on its local data slice, the server aggregates with FedAvg, check that accuracy improves without raw patient images leaving any node	◦ FL PoC running on 3 simulated nodes ◦ Report: accuracy comparison, convergence chart, round-trip latency

		• This is a proof-of-concept, not yet deployed to production. Document accuracy vs centralised baseline and round-trip latency. If the accuracy gap is large, explain why and what data volume would close it	
Week 4	**Multimodal AI + clinical NLP**	• The AI models see only the photograph. Build a multimodal prompt for Gemini 1.5 Pro Vision that sends both the photograph and the patient's HbA1c, diabetes duration, and blood pressure together, returning a richer severity assessment as structured JSON. Test on 20 sample cases • Doctors write free-text consultation notes. Build a spaCy pipeline with a custom entity ruler to extract wound_location, infection_sign, and treatment_recommendation. Store the structured output alongside the free text	◦ Multimodal Gemini prompt (tested on 20 cases, structured JSON output) ◦ spaCy NLP pipeline with custom entity ruler ◦ Structured NLP output stored in DB
Week 5	**RAG assistant + consent versioning**	• Build a RAG assistant over the field worker training manual: PDF chunking, embedding, FAISS vector index, LangChain retrieval, exposed as POST /fieldworker/ask. Test with 20 sample questions, at least 18 should return a relevant answer • Consent versioning: two consent stages exist for different data uses. If the data use changes, the system needs to flag which users need to re-consent. Wire through the consent_versions table • Write plain-language consent summaries in two languages. Use an LLM to draft, then read them carefully and verify accuracy yourself	◦ POST /fieldworker/ask endpoint (18+/20 relevant answers) ◦ Consent versioning workflow (both stages live) ◦ Plain-language consent summaries in both languages (manually QA'd)
Week 6	**Audit & close out**	• Encryption audit: all photographs must be AES-256-GCM encrypted at rest. Every API endpoint must be HTTPS only. Spot-check a few photographs in the database to confirm they are not stored in plaintext • Audit trail: every read, write, delete, and login must write a row to audit_logs. Write a test that makes a series of API calls and checks the log, if any action is missing, something is broken • Work through the OWASP Top 10 checklist for the API endpoints. Document gaps honestly. Write the Privacy Impact Assessment for the IEC ethics submission	◦ Encryption audit report ◦ Audit trail completeness test ◦ OWASP Top 10 checklist (gaps documented) ◦ Privacy Impact Assessment document ◦ FL PoC summary for reporting

Deliverables

#	Deliverable	By	Submit where
1	PII field map (26 tables) + DPDP gap analysis	Wk 1	Drive doc
2	Anonymisation module (Python, k≥5 verified)	Wk 2	GitHub
3	Erasure pipeline, all 26 tables, tested and verified	Wk 2	GitHub + test log
4	Data export integrated with anonymisation layer	Wk 2	GitHub
5	FL PoC report (accuracy vs centralised, latency)	Wk 3	Drive doc + GitHub
6	Multimodal Gemini prompt template (20 cases)	Wk 4	GitHub
7	spaCy NLP pipeline with custom entity ruler	Wk 4	GitHub
8	POST /fieldworker/ask endpoint (18+/20 relevant)	Wk 5	GitHub
9	Consent versioning (both stages live)	Wk 5	GitHub
10	Consent summaries in both languages (QA'd)	Wk 5	Drive doc
11	Encryption + audit trail verification reports	Wk 6	Drive docs
12	Privacy Impact Assessment	Wk 6	Drive doc
13	OWASP Top 10 checklist	Wk 6	Drive doc

Weekly Report Template

One page. Submitted every Friday by 6 PM. The PI will respond by Monday. *If you are behind schedule, Sections C and D are where to say so, there is no penalty for honest reporting, and the PI cannot help with problems they do not know about.*

Heath Screening AI Project | Weekly Progress Report	
One page · Friday by 6 PM · Specific numbers and links, not summaries	
Intern name:	
Role:	
Week number and date:	
GitHub username:	
W&B or Drive link this week:	

A. Work completed this week (4–5 bullets)

Example format (replace these with your own answers):

- *Completed SAM2 wound segmentation module. Tested on 50 images from the dataset, IoU 0.73, above the 0.70 target.*
- *Fixed CameraScreen crash on Android API 28 (null permission result handler). Closed GitHub Issue #31.*
- *Wound severity model fine-tuning complete: 78.4% top-1 on validation set. Target was 75%, so we are through.*
- *Deployed POST /fieldworker/ask endpoint. Tested 20 sample questions. 18 of 20 returned relevant answers.*

Your answer:

B. Code and files submitted (exact links, not 'see GitHub')

Example format (replace these with your own answers):

- *GitHub PR #47, SAM2 segmentation module, branch feat/wound-seg: github.com/[repo]/pull/47*
- *Drive, dataset split and class distribution charts: drive.google.com/[link]*
- *W&B run, wound severity model v3 (run ID ws_v3): wandb.ai/[link]*

Your answer:

C. Problems faced (specific, what broke and what you tried)

Example format (replace these with your own answers):

- *SAM2 IoU drops to 0.41 on high-eschar wounds (low contrast). Trying stronger CLAHE pre-processing this week.*
- *Leprosy class has only 143 images after DermNet NZ download. Oversampling to 800 with augmentation, not yet sure if that is enough for 95% sensitivity.*
- *Gemini API rate limit hit during batch testing. Will throttle to smaller batches.*

Your answer:

D. Help needed from PI (who, what, by when)

Example format (replace these with your own answers):

- *Need the dataset download credentials from PI by Monday, wound severity training is blocked.*
- *Need NHA sandbox client_secret by Wednesday, health ID integration (Flow A) is blocked.*
- *Requesting backend engineer code review on database migration script by Tuesday.*

Your answer:

E. Targets for next week (3–4 measurable goals, include numbers)

Example format (replace these with your own answers):

- *Wound severity model to ≥75% top-1 on validation. Export TFLite and ONNX by Friday.*
- *Wire SAM2 output into tissue CNN, POST /infer/wound returns severity grade and tissue type in one response.*
- *Close all 7 Functional-severity mobile bugs in GitHub Issues by Thursday.*

Your answer:

Self-assessment: On track Slightly behind Need urgent help (circle one)

PI notes: *(returned by Monday)*